Name: V Sunil Kumar

Reg.No:192425292

1. Scenario: You are working on a project that involves analyzing student performance data for a

class of 32 students. The data is stored in a NumPy array named student scores, where each row

represents a student and each column represents a different subject. The subjects are arranged in the

following order: Math, Science, English, and History. Your task is to calculate the average score for

each subject and identify the subject with the highest average score.

Question: How would you use NumPy arrays to calculate the average score for each subject and

determine the subject with the highest average score? Assume 4x4 matrix that stores marks of each

student in given order.

Code;

```
import numpy as np


student_scores = np.array([
   [80, 75, 85, 70],
   [90, 85, 80, 75],
   [70, 80, 90, 85],
   [85, 90, 75, 80]
])


avg = np.mean(student_scores, axis=0)


subjects = ["Math", "Science", "English", "History"]
```

print("Average scores:")

for i in range(4):

  print(subjects[i], ":", avg[i])

highest = np.argmax(avg)

print("Highest average subject:", subjects[highest])

print("Highest average score:", avg [highest])

exp 1 dsa.py - C:/Users/Jashw/exp 1 dsa.py (3.13.14)

File Edit Format Run Options Window Help

```
import numpy as np

student_scores = np.array([
    [80, 75, 85, 70],
    [90, 85, 80, 75],
    [70, 80, 90, 85],
    [85, 90, 75, 80]
])

avg = np.mean(student_scores, axis=0)

subjects = ["Math", "Science", "English", "History"]

print("Average scores:")

for i in range(4):
    print(subjects[i], ":", avg[i])

highest = np.argmax(avg)

print("Highest average subject:", subjects[highest])
print("Highest average score:", avg[highest])
```

output;

```
ile  Edit  Shell  Debug  Options  Window  Help
   Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
   Enter "help" below or click "Help" above for more information.
>> import numpy as np
.. print(np.__version__)
   SyntaxError: multiple statements found while compiling a single statement
>>
   ========================= RESTART: C:/Users/Jashw/vv.py =========================
   2.5.3
>>
   ====================== RESTART: C:/Users/Jashw/exp 1 dsa.py ======================
   Average scores:
   Math : 81.25
   Science : 82.5
   English : 82.5
   History : 77.5
   Highest average subject: Science
   Highest average score: 82.5
```